

Worksheet 2

Exercise 1 On a one-way road, the flow of cars can be described by a Poisson process of intensity $\lambda = \frac{1}{6} s^{-1}$. A pedestrian wishing to cross the road needs an interval of at least 4s between 2 successive cars. Calculate

1. the probability that he will have to wait ;
2. the mean duration of the intervals that allow it to cross the road.

Exercise 2 Calculate the n -th arrival time law of a Poisson process in two ways:

- a) Using $\{S_n \leq t\} = \{N_t \geq n\}$;
- b) Using $S_n = T_1 + T_2 + \dots + T_n$.

Exercise 3 Let a birth and death process with transition rates whose transition rates are:

- a) $\lambda_n = \lambda q$ and $\mu_n = \mu$ s.t. $0 < q \leq 1, \lambda > 0, \mu > 0, n \geq 0$.
- b) $\lambda_n = \frac{\lambda}{n+1}$ and $\mu_n = \mu$ s.t. $\lambda > 0, \mu > 0, n \geq 0$.
- c) $\lambda_n = \lambda$ and $\mu_n = \mu$ s.t. $\lambda > 0, \mu > 0, \mu_0 = 0, n \geq 0$.

Determine stationary distribution of the process in question

Exercise 4 A curious resident is interested in the phone booth located at the bottom of his building. After several months of persistent observation, he is able to state that the arrival of people coming to make phone calls follows a Poisson process with an arrival rate of 5 people per hour. He has also calculated that the duration of a phone call corresponds to an exponential random variable with a mean of 10 minutes.

1. On average, how many people are waiting in front of the booth?
2. What is the average waiting time in front of the booth?
3. The telephone company wants to add a second booth in locations where the existing one is occupied more than 75 percent of the time. Should they add one in front of this building?

Exercise 5 Find the distributions of a customer's waiting time in the queue and of a customer's sojourn time in the M/M/1 system. Prove the same result for the sojourn time, however, using the Laplace–Stieltjes transform.

Exercise 6 A public agency is open every working day from 9 am to 5 pm without interruption. It welcomes a mean of 64 users a day; one counter is used solely to process each user's file. A statistical study has shown that the random duration of services follows an exponential distribution with a mean of 2.5 minutes, and that user arrivals form a Poisson process. It is assumed that steady state is rapidly reached.

1. Give the Kendall notation for this queuing system; the mean time spent waiting; the mean time spent in the organization for each user.
2. What are the probabilities that no customers will arrive between 3pm and 4pm? That 6 customers arrive between 4pm and 5pm?
3. What is the probability of observing a queue of 4 customers, behind the current customer?
4. What is the probability that a customer will spend more than 15 minutes in the organization?

Exercise 7 Customers' arrivals at a bank follow a Poisson process with a rate of 9 customers per hour. The service time per customer is considered as an exponentially distributed random variable with a mean of 10 minutes. What is the minimum number of service counters required to ensure a steady state of this process? What is the average waiting time if the number of counters is $c = a$ and $c = a + 1$?

1. Calculate $E[Y]$, where Y is the number of idle servers in an $M/M/c$ system in terms of the overall traffic intensity. Does the average number of busy servers depend on c ?

Exercise 8 A bankruptcy trustee's office employs three analysts at its head office. It is estimated that individuals go bankrupt purely at random, so we can assume that they arrive at the trustee's office following a Poisson process, at a rate of 20 individuals per 8-hour day. The time an analyst spends with a client follows an exponential distribution with a mean of 40 minutes. Clients are served on a first-come, first-served basis.

1. Find the proportion of time an analyst spends interviewing clients.
2. What is the probability that there will be no clients in the trustee's office?
3. What is the average time a client spends in the system?

Exercise 9 Consider the $M/M/2/4$ queueing system. Assume that it is in a steady state; the state probabilities are $1/16; 4/16; 6/16; 4/16$, and $1/16$ (respectively for system states: 0; 1; 2; 3; 4).

1. Calculate the mean number of customers in the system and the mean number of customers in the queue.
2. Assuming that customers arrive in the system according to a Poisson process with $\lambda = 2$ customers/hour, calculate the mean waiting time for a customer. Estimate the mean service time for a customer.

Exercise 10 A workshop has 5 machines (sources of potential customers), each one having a Poissonian breakdown rate equal to $1/6$ (per hour). Two mechanics are in charge of maintenance: the average repair time is 4 hours, following an exponential law.

1. Identify the model using Kendall's notation. Model the problem using an appropriate stochastic process. What is its state space?
2. Draw a simplified transition graph. Describe the transient regime and obtain the stationary distribution, calculating state probabilities.
3. Find the mean number of breakdowns and the mean number of machines in the repair queue. What is the probability of a repairer being active?